

Lecture 12: Alignment Loss

Definition: Let B be a set of gold blocks and P be predicted blocks.

The alignment loss is zero only when every required block is matched once.

Theorem 1

If every block in B has a unique exact textual match in P and the reading order is preserved, then the block-order penalty is zero.

Proof: The monotone alignment maps each gold index to the same predicted index order.

Equations

$$L = 1 - \frac{|M|}{|B|}$$

$$R = \frac{2PR}{P+R}$$

where M is the matched block set, P is precision, and R is recall.